Design_Doc_WWD

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by **Anders Bandt**

Project Abstract

Scope: *This project will attempt to create an electronic wearable, worn on the wrist. The main function of this wearable will be to gather biometric data about the body using optical sensors. Specifically, using green LEDs and photodiodes.*

The TI integrated circuit AFE4432 will be used as the analog front-end for sampling the optical sensors and handling the signal chain.

Photoplethysmography will be used as the technique for extracting volumetric changes in blood in peripheral circulation from the sensor data. From this volumetric blood data, an estimate of heart rate will be produced.

An IMU will be included on the device for incorporating motion artifacts as error into the signal chain.

Bluetooth connectivity and USB are included on the device for data extraction.

Device Components

MCU
Using the TI MCU CC2642R from the SimpleLink portfolio.
This is a 32-bit ARM Cortex-M4F wireless MCU with Bluetooth capabilities.

AFE
The AFE4432 - WWD chip from TI will be used as the analog front end. Internal AFE clocking will be used, with flexibility to use single-shot clocking through a GPIO pin on the MCU.

There was a mistake in layout and currently only I2C capability is built into the design. Additionally, the pull-up resistors are pulled up to **TX_SUP** instead of **RX_SUP**, which will require some creative wire rework.

SPI capability is still possible, but would require some rework (one wire soldered onto PCB)

Connectivity

1. USB
    1. Enabled by the FTDI chip FT230X
    2. Converts USB data into UART for the MCU
2. Bluetooth
    1. 2.4 GHz chip antenna included in the design, feeding through a matching network into the MCU
3. JTAG - Debug
    1. cJTAG and JTAG exposed through conenctor

Sensors

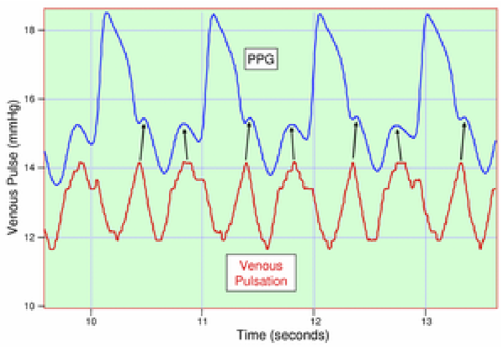
1. 4 x green LED emitters
2. 3 x photodiodes
3. IMU
    - ICM-42670 (IMU) - WWD
    - 3-axis gyro and accelerometer

Background and Theory

Below is an explanation of the operating principle behind heart rate extraction, photoplethysmography.

PPG Explanation

A **photoplethysmogram (PPG)** is an optically obtained plethysmogram that can be used to detect blood volume changes in the microvascular bed of tissue. With each cardiac cycle, the heart pumps blood to the periphery. The change in volume caused by hte pressure pulse is detected by illuminating the skin with the light from an LED, and then measuring the amount of light either transmitted or reflected to a photodiode.



venous pulsation clearly seen in a PPG

Each cardiac cycle will appear as a peak, heart rate can be determined. The DC component of the signal is attributed to the bulk absorprtion of the skin tissue, and should be canceled out in signal processing.

The AC signal is directly attributable to the variation in blood volume in the skin caused by the pressure pulse of the cardiac cycle. The height of this AC signal is the proportional to pulse pressure (the difference between the systolic and diastolic pressure in the arteries). This is the true "signal of interest".

Signal Chain

There are many various components but much of the analog side of it is handled by the AFE4432. It seems to be that there will be a couple set of challenges.

- **Noise**
  - Various sources of noise in system (sensitivity in power supplies,)
- **Filtering**
  - Progammable filters on AFE
  - Filtering methods applied to digital FIFO data to improve signal
- **Timing**
  - Complex phase timing sets
  - Many options configurable on AFE

The details of the signal chain are not fully fleshed out at this point, but below is a rough outline of some various sub-blocks of the signal chain.

Sub-blocks of Signal Chain

Sensor Input

The signal chain starts with the sensors: photodiodes and LEDs. The AFE has two LED drivers each with 8-bit current control.

Low noise offset DACs can be enabled at the sensor inputs to automatically cancel DC.

Amplification

The raw sensor signals are typically weak and require amplification to increase the signal level for further processing. The AFE4432 has a low-noise programmable gain amplifier (PGA) that can be configured to amplify the sensor signals according to the application requirements.

Up to 12 signal phase sets can be defined, each phase set comprising a combination of LED and Ambient phases.

The current from the photodiodes in each phase is converted into a voltage by a TIA, which is then fed into filtering.

Filtering

After amplification, filtering is applied to remove unwanted noise and interference from the sensor signals. The AFE4432 includes a digital finite impulse response (FIR) filter with programmable coefficients, which can be used to implement different filter types such as low-pass, high-pass, band-pass, or notch filters to achieve the desired signal quality.

Decimation

In some cases, the sensor signals may be oversampled, and decimation can be applied to reduce the data rate and improve the signal-to-noise ratio. The AFE4432 includes a digital decimation filter that can be used to downsample the sensor signals while preserving the relevant frequency content.

ADC Sampling

The AFE4432 features an integrated ADC that can be used to sample the filtered and decimated sensor signals. The sampled signals are stored in a 160-sample FIFO block, which will be read out to the MCU through either a SPI or I2C interface.

DSP

The digitized sensor signals will undergo various digital signal processing (DSP) techniques, such as filtering, feature extraction, and analysis, depending on the specific application requirements.

Heart rate detection will require some digital algorithms. Similiar to R-wave detection in ECG QRS complex processing, the algorithm will mainly need to detect the relative peak of the PPG signal and compute the time difference from peaks, finding frequency of peaks. The detail of this signal processing isn't fully mapped out yet, but some potential options include

- **Fast Fourier Transform (FFT)**
  - Converts the PPGG signal from the time domain to the frequency domain
- **Autocorrelation function (ACF)**
  - Measures the similarity between a signal and a delayed verison of itself
- **Wavelet transform**
  - Used to analyze signals with both frequency and time variations.
  - Can provide detailed information about the PPG signal at different time scales, and from that used to extract heart rate information from the PPG signal

Phase Cancellation (Optional)

In some applications, phase cancellation techniques may be applied to remove artifacts or interference from the sensor signals. This could involve generating a reference signal with an opposite phase to the interference and subtracting it from the sensor signal to cancel out the unwanted component. The specific implementation of phase cancellation would depend on the type of interference and the desired signal quality.

Output

Finally, the processed sensor signals can be output to a display, storage device, or transmitted wirelessly for further analysis or display on a user interface, depending on the application requirements.

For this initial design, the data will be either output through Bluetooth or USB.

Power System

Regulators

BMS - BQ25120A

The BMS will be done through the TI chip bq25120A - WWD. This IC is highly integrated, with a multitude of features.

Specifically used in this design will be

- Linear charger
- Unregulated buck output (PMID)
- Battery temperature monitoring (through thermistor)

Boost Converter - TPS6125

In order to get a higher voltage rail in the system, a boost converter will be used. It will be fed from the buck (PMID) output from the BMS

AFE4432 Supplies

LDO - TX_SUP

The AFE4432 has a transmit supply, mainly used for driving the LEDs. For this, the LDO LP38693 will be used with an output of 3.3V

LDO - RX_SUP

The AFE4432 also has a receive supply, used for the photodiodes and other IC power. For this, the LDO LP5900 - WWD will be used at an output of 1.8V.

Voltage and Current Ratings

Operating Voltages
The table below lists the "desired voltages", meaning the voltages intended during normal operation of the device

> Ideal Operating Voltages
>
> | Line | Min | Typ | Max | Spec Driver |
> |---|---|---|---|---|
> | RX_SUP | 1.7 | 1.8 | 1.9 | AFE4432 6.3 - Recommended Operating Conditions |
> | TX_SUP | 3.0 | 3.3 | 3.6 | AFE4432 6.3 - Recommended Operating Conditions |
> | VDDS | 1.96 | 2.6 | 3.0 | See Note 1 |
> | BOOST | 4.8 | 5 | 5.2 | TPS61252 - WWD > Ideal Output Voltage |
> | PMID | 2.7 | 3.3 | 5.0 | ideal voltage to maintain VDDS |
> | CHARGE_IN | 4.8 | 5.0 | 5.2 | gut guess if powered through USB |
> | BAT_+ | 2.85 | 3.6 | 4.20 | I made up 3.6 V as typical/ideal |
>
> *Note 1: Upper limit driven by bq25120A - WWD VSYS limits with an applied margin of (-0.3 V). Lower limit driven by my calculations in bq25120A - WWD > Operating Parameters*

Operating Currents
The table below lists the maximum current ratings

> Maximum Current
>
> | Line | Min | Typ | Max | Spec Driver |
> |---|---|---|---|---|
> | RX_SUP | | | 3 mA | |
> | TX_SUP | | | 200 mA | LED forward current, AFE4432 consumption (note 1) |
> | VDD | | | 15 mA | Not sure where I got this |
> | PMID | | | 300 mA | BMS integrated BUCK |
> | CHARGE_IN | | | | |
> | BAT_+ | | | 450 mA | |
>
> **Notes**

1. Depending on my max LED forward current and the AFE4432 device itself current draw?
    1. 15mA * 4 = 60 mA

Absolute Maximum Voltages

Below is the list of the absolute maximum votlages that should be on any pin.

Absolute Maximum Voltage

Line	Min	Typ	Max	Spec Driver
RX_SUP	-0.3		2.1	AFE4432 6.1 - Absolute Maximum Ratings
TX_SUP	-0.3		6	AFE4432 6.1 - Absolute Maximum Ratings
VDDS	-0.3	2.6	3.6	IMU (CC2642R is max 4.1 V)
BOOST	?		?	TPS61252 Max Ratings
PMID	1.8		5.0	BMS integrated BUCK
CHARGE_IN			8.0	
BAT_+	2.75		4.25	

Physical Assembly, Casing

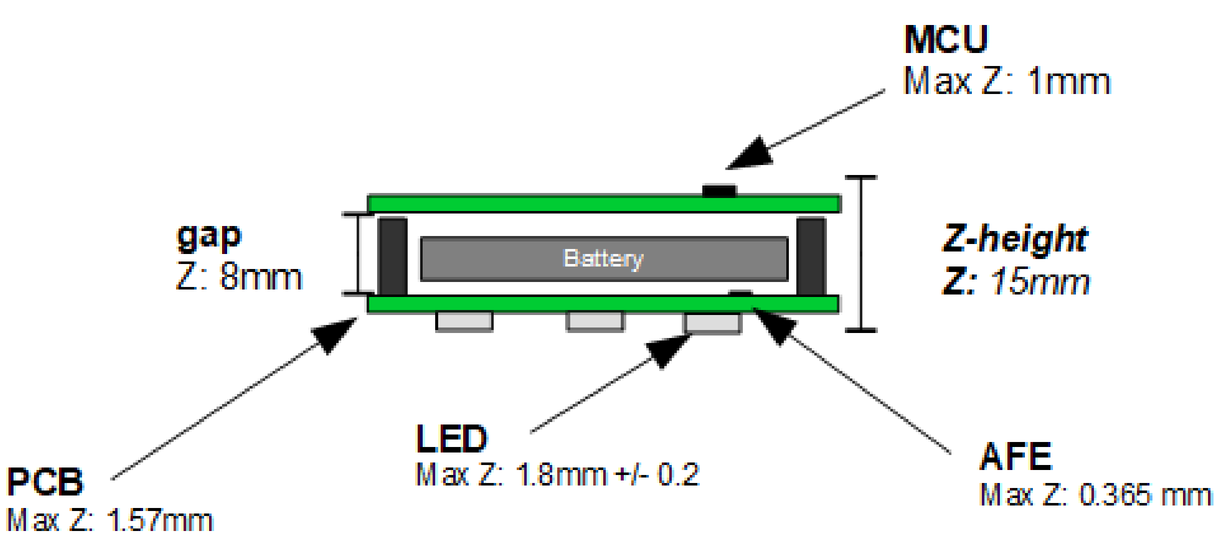
The design will be separated into two boards.

Board 1 (bottom board) shall contain:

1. AFE4432
2. LEDs
3. Photodiodes

Board 2 (top board) shall contain:

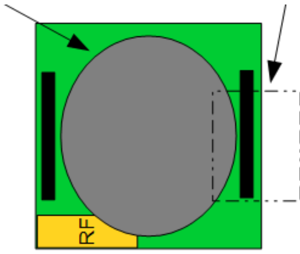
1. MCU
2. RF Antenna
3. Power system



side view of board

This separation of the boards was done for a couple reasons.

- Better signal integrity on the LED/photodiode side.
    - Less interference from switching buck/boost converters, RF, etc.
- Modularity of design
    - Allows for better testability of the MCU and AFE separately



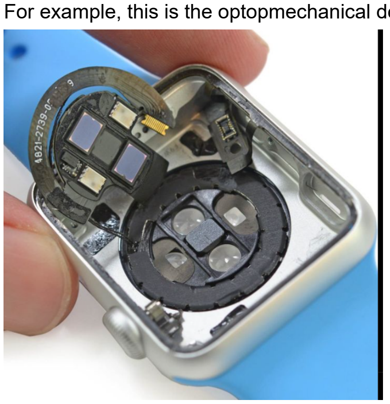
XY Plane: 45x45 mm

top down view of the XY plane. Board dimensions are 45 x 45 mm

Female headers (Samtec SSM-110-L-SV) will be mounted on the bottom board, with mating connectors (Samtec MTLW series) coming down from the top.

Casing

Casing is undetermined at this point. Ideally, casing will increase signal integrity of the LED/photodiode signals. This means a few things

1. Small curvature in the casing to allow depression into the worst, and better optical coupling (less ambient light leakage as well)
2. Lenses to focus emitted and reflected light
    1. For example, this is the optopmechanical design on the casing of an Apple Watch
    
    2.

Testing

After my initial year working as a product engineer with Texas Instruments, I have developed some appreciation for quality testing procedures. And from that experience I have developed some testing procedures to "wakeup" the device.

Board 1
Below is the procedure to "wakeup" b1 (AFE4432)

> **B1 Wakeup - AFE4432**
>
> ---
>
> [AFE4432 - WWD](AFE4432 - WWD)
> Wakeup on board 1

> Initialization Procedure
>
> Start with AFE4432 board, applying power externally and issuing simple commands
>
> Process should be like so
>
> 1. Apply power, if current draw good and no shorts can proceed
>   1. Consideration - AFE4432 has some power supply ramping requirements for TX_SUP and RX_SUP
> 2. Apply a reset
>   1. Done by writing to a `SW_RESET` register bit
>   2. Can also be a hardware reset.... details pending???
> 3. Write `PAGE_SEL` to 1
>   1. AFE Firmware - WWD > Page Selection
> 4. Write all the relevant register settings in Page 1 of the register map
> 5. Write `PAGE_SEL` to 1
> 6. Program the registers in Page 0. Registers can be programmed in any order
> 7. At this point, can enable ISR in the MCU from the AFE
> 8. Set the `TIMER_ENABLE` and `PRF_COUNTER_ENABLE` bits to 1. Timing engine will start running and the FIFO will start off in an initialized state

Board 2
And below is the procedure to "wake up" board 2, containing the power system and MCU.

> **B2 Wakeup - MCU**
>
> ---
>
> MCU - WWD > MCU Wakeup
>
> MCU wakeup will involve
>
> 1. Observing valid MCU power
> 2. Uploading program
> 3. Blinking LED
>
> > **MCU Wakeup**
> >
> > ---
> >
> > Here is the procedure for waking up the MCU
> >
> > 1. Apply ideal voltage to **BAT_+**
> >   1. Observe 1.8V on **VDDS**
> >   2. This is Vmin for the MCU so should be able to issue commands
> > 2. Apply system power (VDDS settings) to BMS over I2C
> > 3. Observe 2.6 V on **VDDS**
> > 4. Force power-on reset condition on the CC2642R
> >   1. **MCU_RESET** should be held high
> >   2. Need to hold low for at least $1\mu s$
> > 5. Start bootloader (should auto start with no valid flash image)
> > 6. Upload blink example using `LED_STATUS` (DIO_0) to the MCU
> > 7.
> >
> > Good luck have fun!

Hard problems to deal with

There are many challenges with this project. It's hard to say that the hardware I designed will work fully, or at all.

The signal chain will be very sensitive to any noise and interference. Hopefully the AFE will be intuitive to use, but layout and casing will also be important factors in ensuring a quality signal chain.

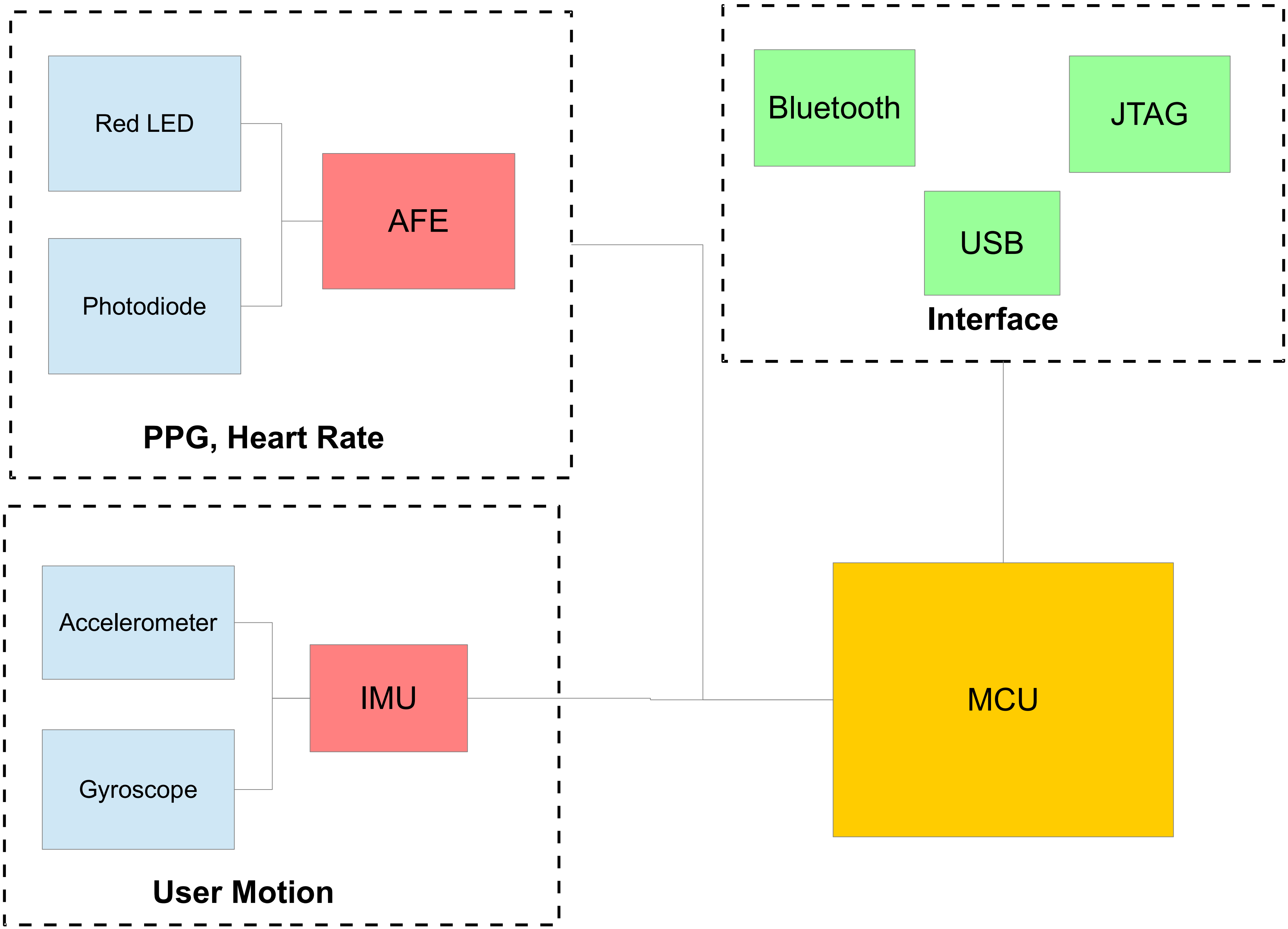
The digital processing will require a large software effort to fully process the data.

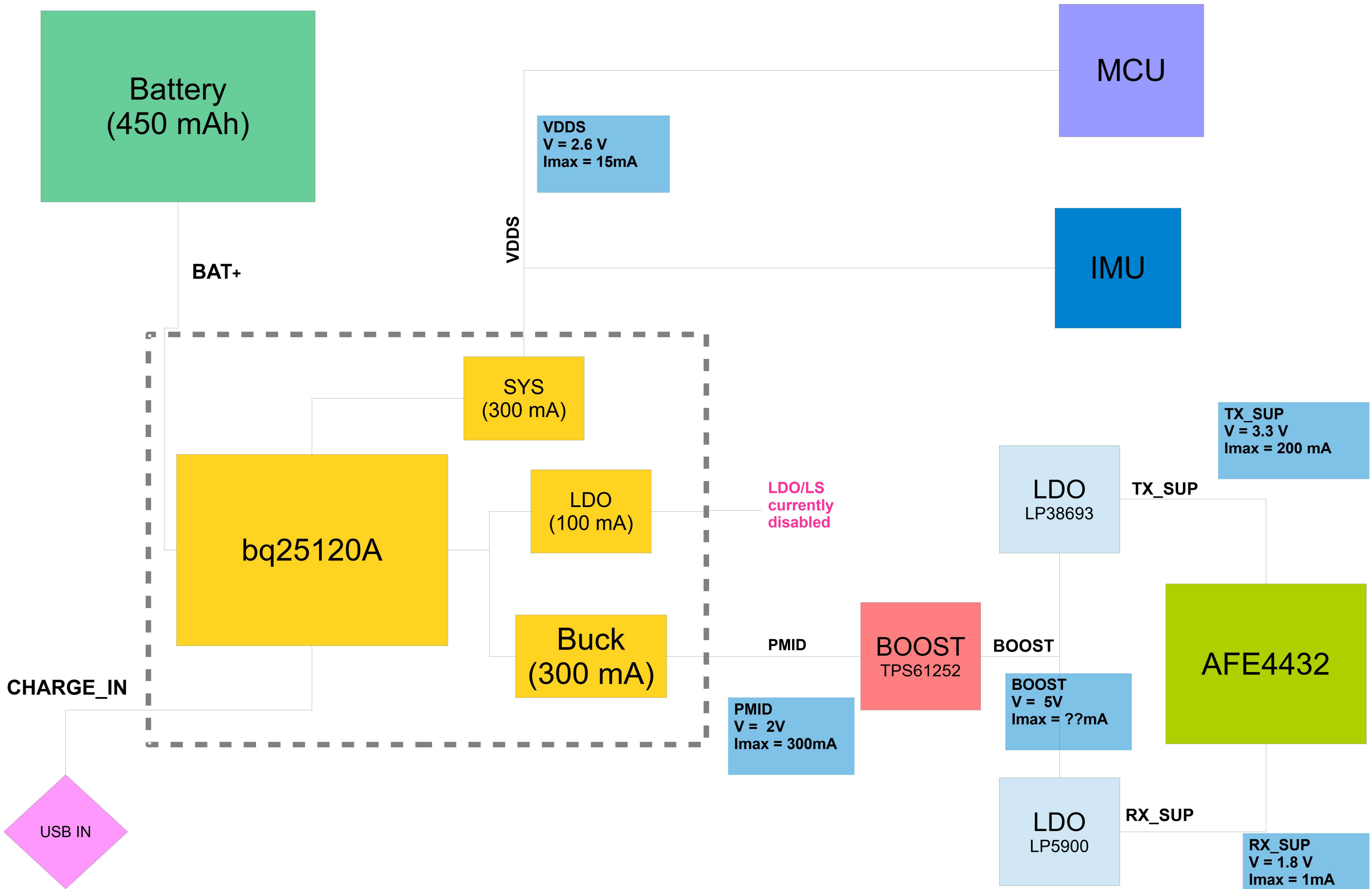
Optomechanic Challenges

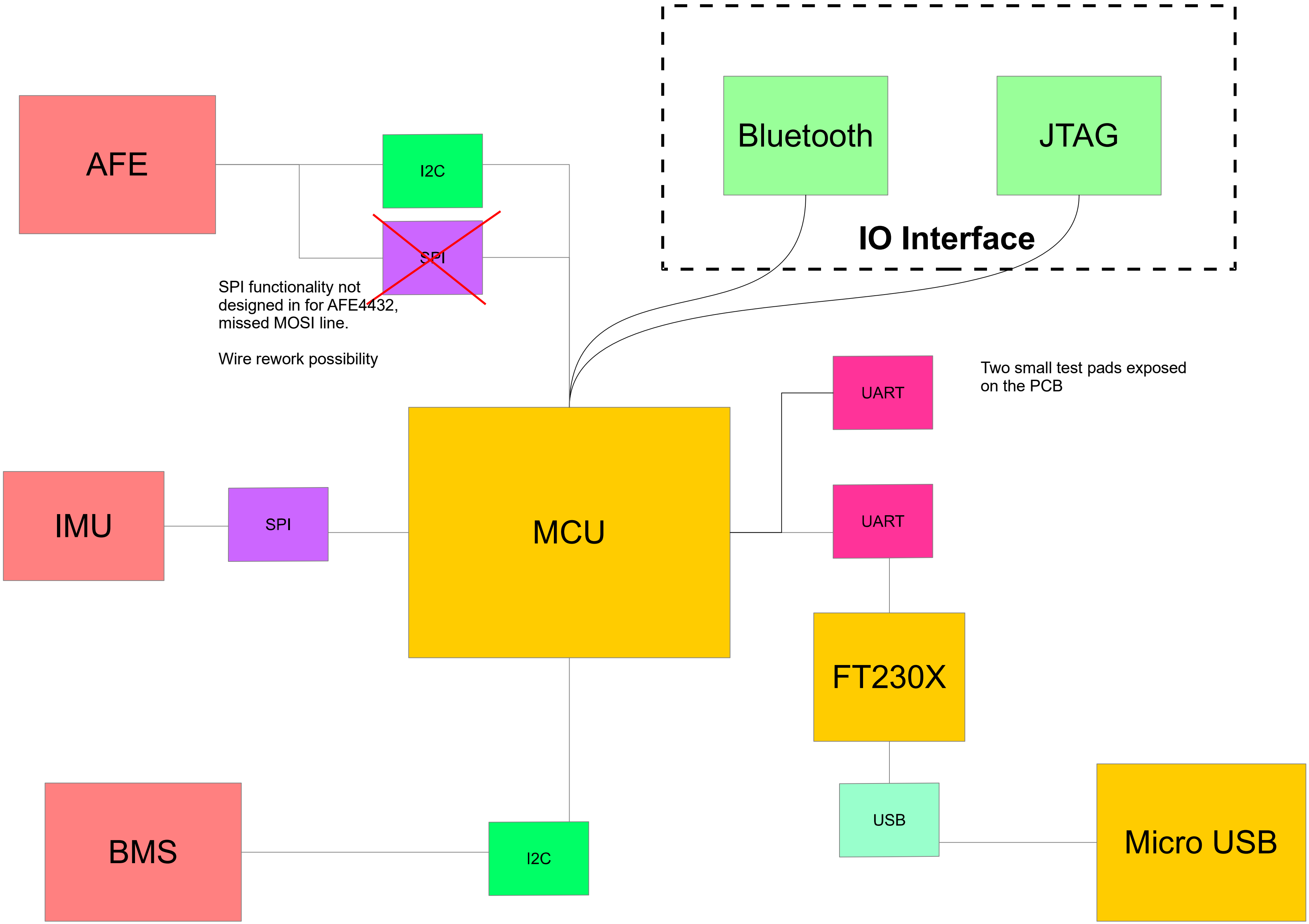
- Coupling
- Wavelengths
- Multiple form factors
- Multiple emitters
- Gross displacement

Signal Extraction Challenges

- Motion-tolerant
- Validation
- Performance
- Power management
- Scalable biometric roadmap


Red LED
Photodiode
AFE
PPG, Heart Rate
Bluetooth
JTAG
USB
Interface
Accelerometer
Gyroscope
IMU
User Motion
MCU


Battery
(450 mAh)
BAT+
VDDS
VDDS
V = 2.6 V
Imax = 15mA
MCU
IMU
SYS
(300 mA)
bq25120A
LDO
(100 mA)
Buck
(300 mA)
LDO/LS
currently
disabled
CHARGE_IN
USB IN
PMID
PMID
V = 2V
Imax = 300mA
BOOST
TPS61252
BOOST
BOOST
V = 5V
Imax = ??mA
LDO
LP38693
TX_SUP
TX_SUP
V = 3.3 V
Imax = 200 mA
AFE4432
LDO
LP5900
RX_SUP
RX_SUP
V = 1.8 V
Imax = 1mA


AFE
I2C
SPI
SPI functionality not designed in for AFE4432, missed MOSI line.
Wire rework possibility
Bluetooth
JTAG
IO Interface
IMU
SPI
MCU
UART
Two small test pads exposed on the PCB
UART
FT230X
BMS
I2C
USB
Micro USB

Wearable Wrist Device

Board Dimensions

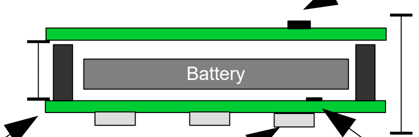
Connector Edge Clearance

Clearance (C) = ((pcb_x – bat_x)/2 – con_x)/2
C = 1.23 mm

Where
pcb_x = 45 mm
bat_x = 35 mm
con_x = 2.54mm

MCU
Max Z: 1mm

gap
Z: 8mm



Z-height
Z: *15mm*

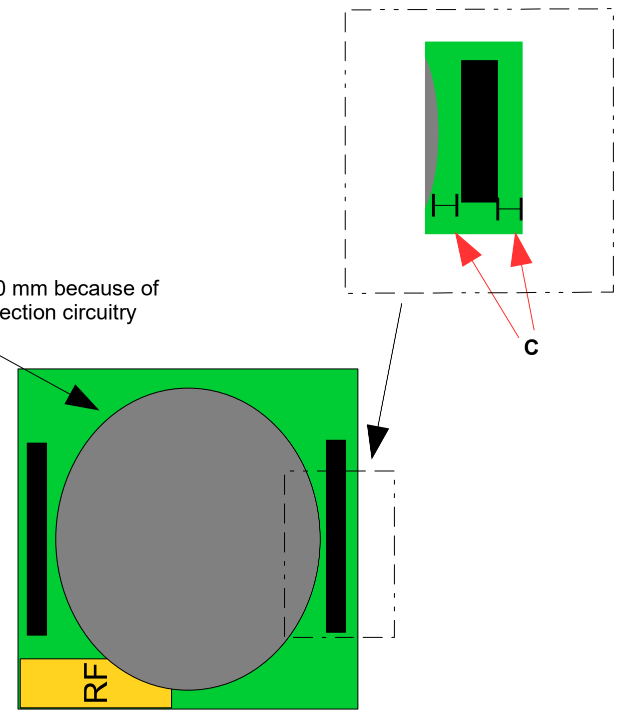
PCB
Max Z: 1.57mm

LED
Max Z: 1.8mm +/- 0.2

AFE
Max Z: 0.365 mm

Note: not sure where that number came from
Altium says 16.2 mils (.42 mm)

Note
made battery y axis 40 mm because of
the added battery protection circuitry



Z Plane: 15 mm

XY Plane: 45x45 mm

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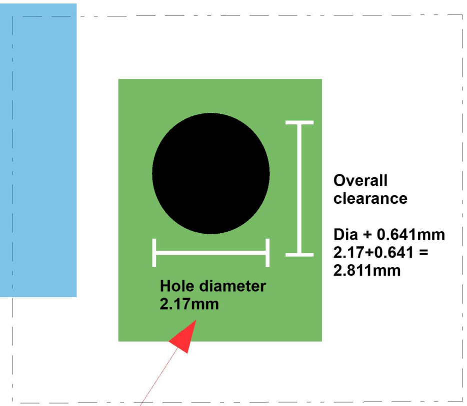
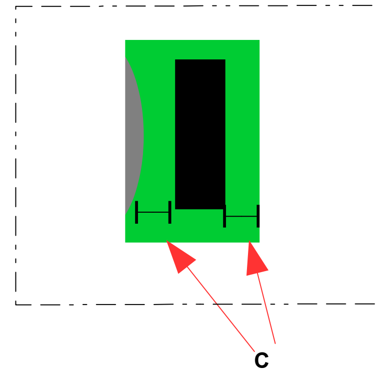
Wearable Wrist Device

Connector and Mounting Holes





Dimensioning Note 1 – Connector

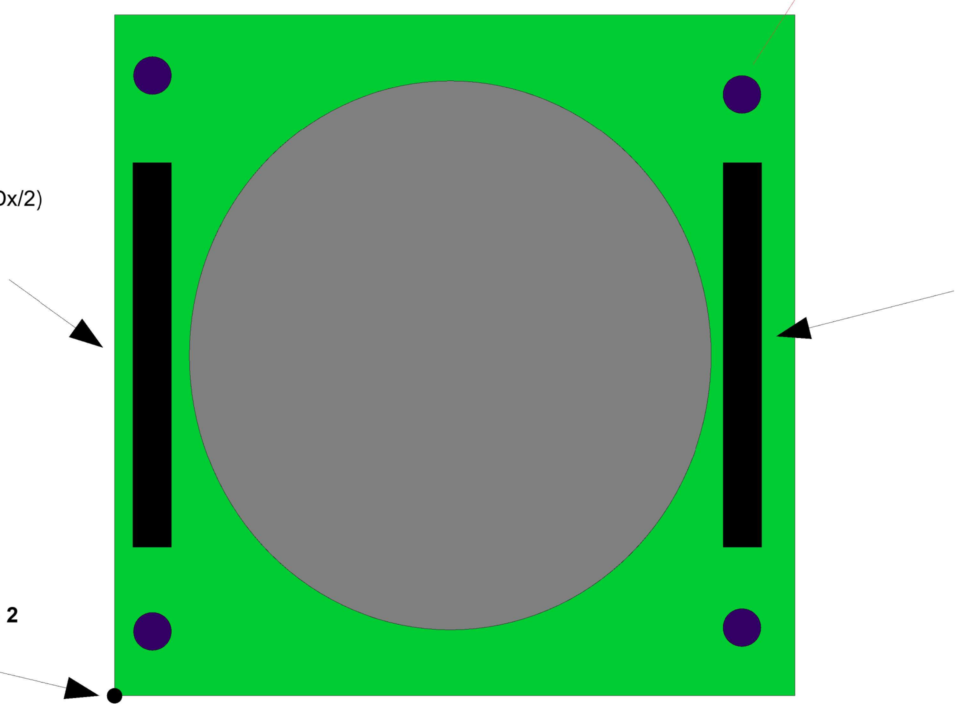
w1 = 2.54mm (width of connector)
Dx = 45mm (length of board in x axis)
C = 1.23mm (connector clearance)

J1

Geometric (x, y)
(x, y) = (x0 + C + w1/2, Dx/2)
= (2.5, 22.5) mm

P2

(x, y) = (4.0, 22.5) mm



J2

Geometric (x, y)
(x, y) = (x1 – C – w1/2, Dx/2)
= (42.5, 22.5)

P2

(x, y) = (41.0, 22.5) mm

Dimensioning Note 2
Origin
(x, y) = 0

Notes
- Dimensions are exactly doubled on this sheet
- Made battery y axis 45 mm because of the added circuitry

XY Plane: 45x45 mm

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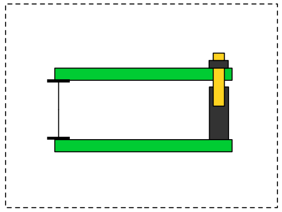
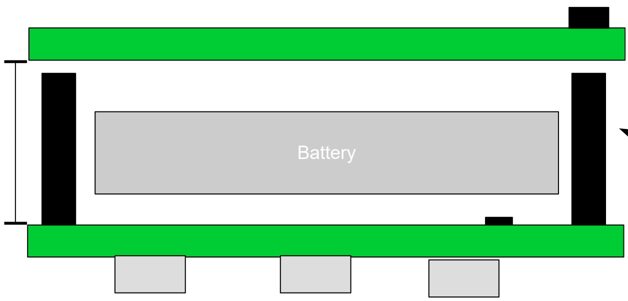
Wearable Wrist Device

Connect Mating

Mating Plunge Depth
Max Z:

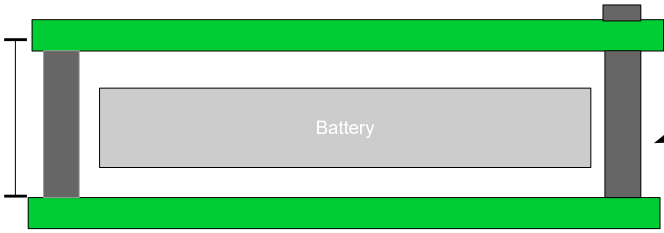
gap
Z: 8mm





SSM
Max Z: 7.37mm

Connector Mating Depths



Between board height: 8mm

Standoffs

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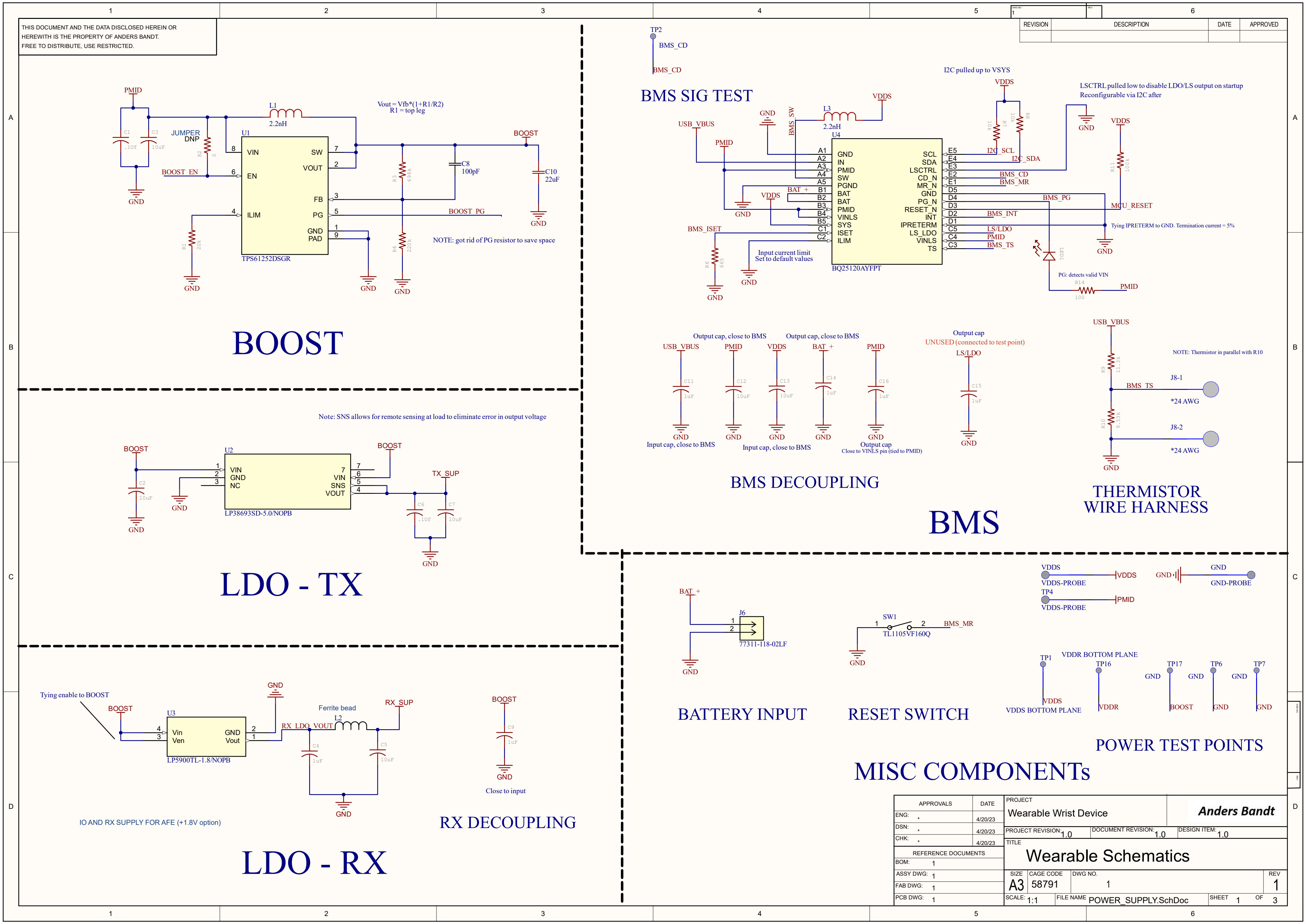

APPROVALS		DATE
ENG:	*	4/20/23
DSN:	*	4/20/23
CHK:	*	4/20/23

REFERENCE DOCUMENTS	
BOM:	1
ASSY DWG:	1
FAB DWG:	1
PCB DWG:	1

PROJECT: Wearable Wrist Device

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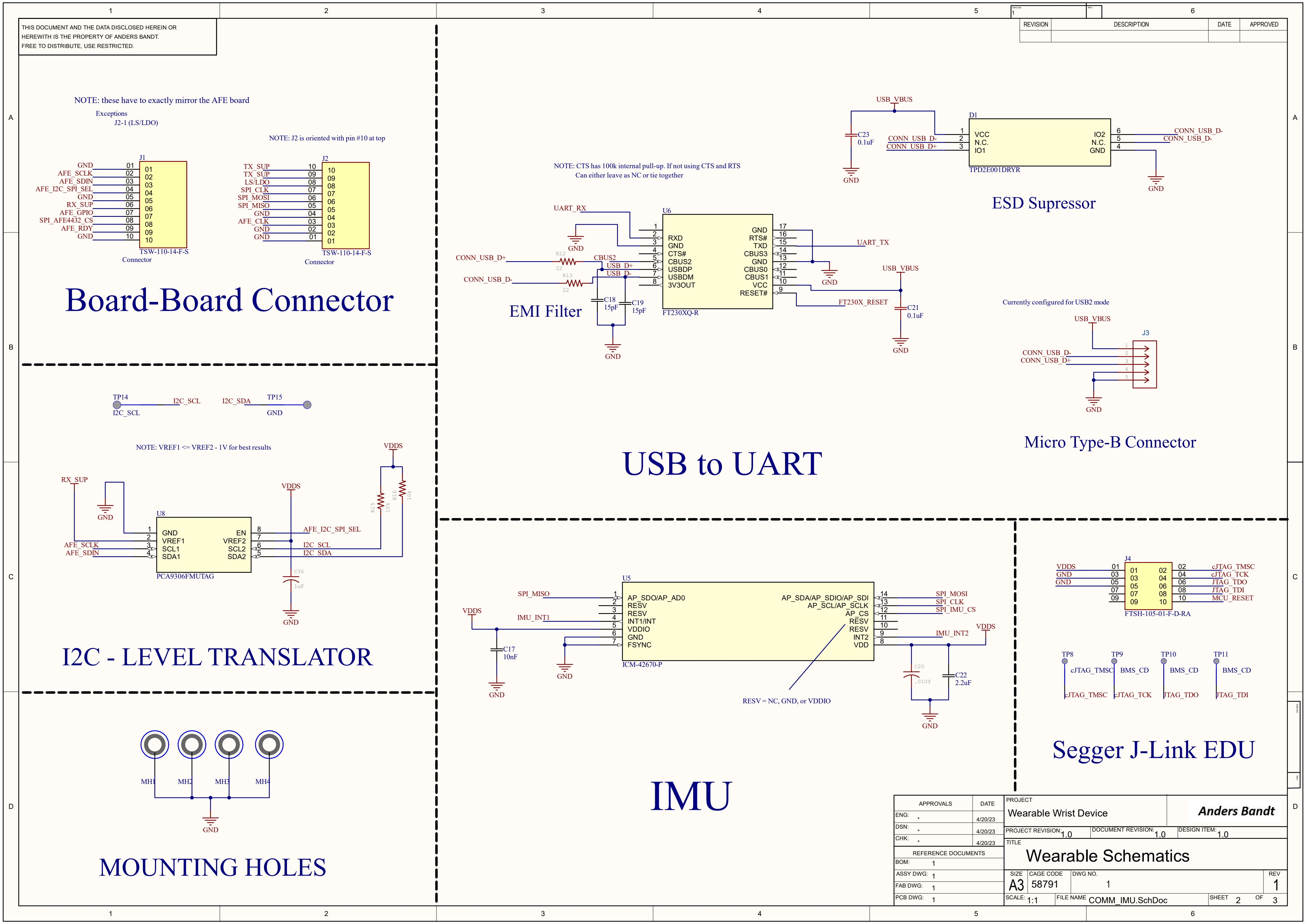
PROJECT REVISION: 1.0 | DOCUMENT REVISION: 1.0 | DESIGN ITEM: 1.0

TITLE: Wearable Schematics

SIZE: A3 | CAGE CODE: 58791 | DWG NO.: 1 | REV: 1

SCALE: 1:1 | FILE NAME: POWER_SUPPLY.SchDoc | SHEET 1 OF 3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REVISION
DESCRIPTION
DATE
APPROVED
NOTE: these have to exactly mirror the AFE board
Exceptions
J2-1 (LS/LDO)
NOTE: J2 is oriented with pin #10 at top
J1
GND
AFE_SCLK
AFE_SDIN
AFE_I2C_SPI_SEL
GND
RX_SUP
AFE_GPIO
SPI_AFE4432_CS
AFE_RDY
GND
TSW-110-14-F-S
Connector
J2
TX_SUP
TX_SUP
LS/LDO
SPI_CLK
SPI_MOSI
SPI_MISO
GND
AFE_CLK
GND
GND
TSW-110-14-F-S
Connector
Board-Board Connector
NOTE: CTS has 100k internal pull-up. If not using CTS and RTS
Can either leave as NC or tie together
UART_RX
U6
RXD
GND
CTS#
CBUS2
USBDP
USBDM
3V3OUT
GND
RTS#
TXD
CBUS3
GND
CBUS0
CBUS1
VCC
RESET#
FT230XQ-R
UART_TX
CONN_USB_D+
CONN_USB_D-
CBUS2
USB_D+
USB_D-
R12
R13
C18 15pF
C19 15pF
USB_VBUS
C21 0.1uF
FT230X_RESET
EMI Filter
GND
USB_VBUS
C23 0.1uF
D1
VCC
N.C.
IO1
IO2
N.C.
GND
CONN_USB_D-
CONN_USB_D+
TPD2E001DRYR
ESD Supressor
Currently configured for USB2 mode
USB_VBUS
J3
CONN_USB_D-
CONN_USB_D+
Micro Type-B Connector
USB to UART
TP14
I2C_SCL
I2C_SDA
TP15
NOTE: VREF1 <= VREF2 - 1V for best results
RX_SUP
VDDS
U8
GND
VREF1
SCL1
SDA1
EN
VREF2
SCL2
SDA2
AFE_SCLK
AFE_SDIN
AFE_I2C_SPI_SEL
I2C_SCL
I2C_SDA
R15 10k
R16 10k
C36 1uF
PCA9306FMUTAG
I2C - LEVEL TRANSLATOR
U5
SPI_MISO
IMU_INT1
VDDS
AP_SDO/AP_AD0
RESV
RESV
INT1/INT
VDDIO
GND
FSYNC
AP_SDA/AP_SDIO/AP_SDI
AP_SCL/AP_SCLK
AP_CS
RESV
RESV
INT2
VDD
SPI_MOSI
SPI_CLK
SPI_IMU_CS
IMU_INT2
C17 10nF
C20 .010f
C22 2.2uF
ICM-42670-P
RESV = NC, GND, or VDDIO
IMU
J4
VDDS
GND
GND
cJTAG_TMSC
cJTAG_TCK
JTAG_TDO
JTAG_TDI
MCU_RESET
FTSH-105-01-F-D-RA
TP8
TP9
TP10
TP11
cJTAG_TMSC
cJTAG_TCK
JTAG_TDO
JTAG_TDI
BMS_CD
Segger J-Link EDU
MH1
MH2
MH3
MH4
GND
MOUNTING HOLES
APPROVALS
DATE
ENG: 4/20/23
DSN: 4/20/23
CHK: 4/20/23
REFERENCE DOCUMENTS
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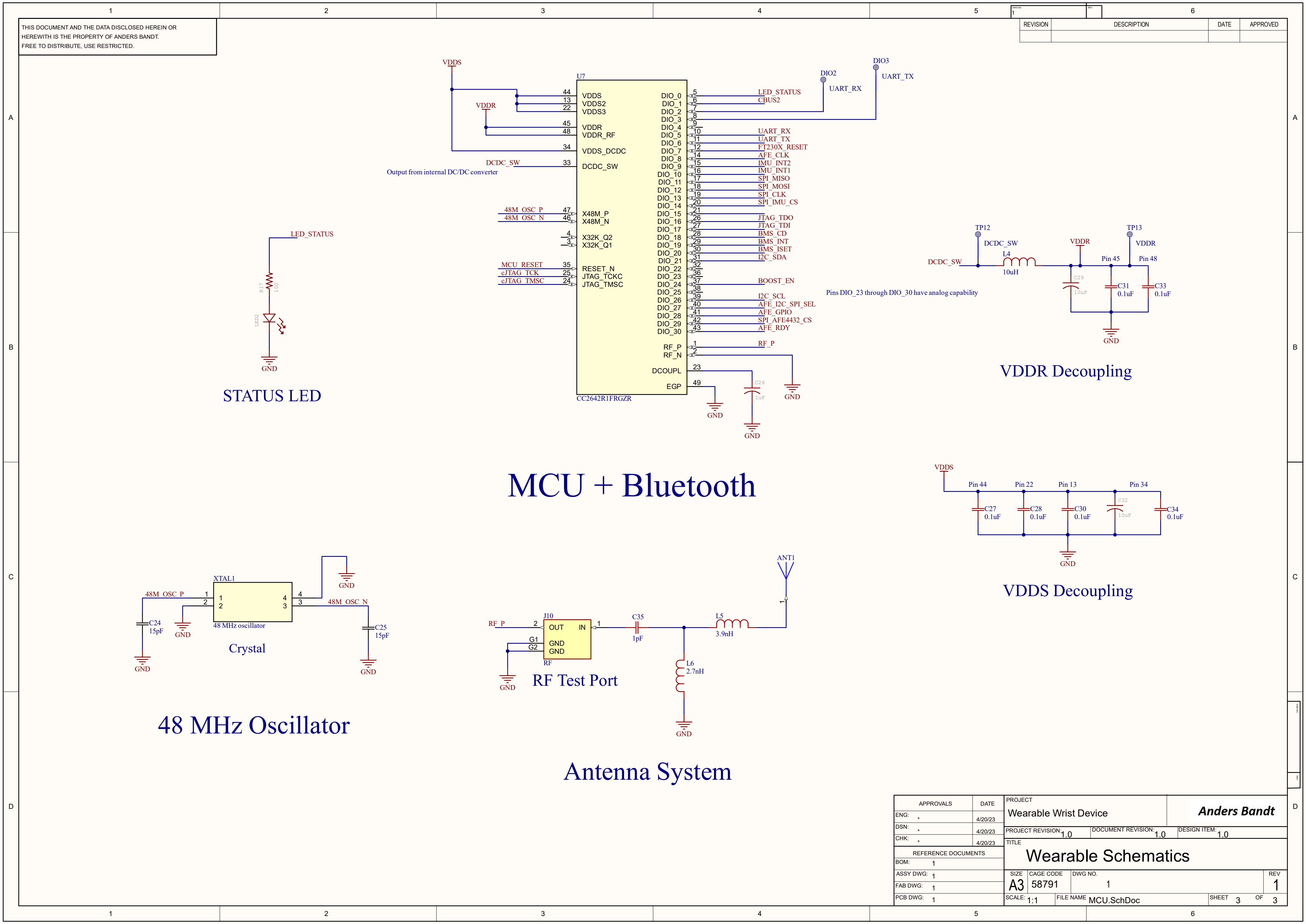
ASSY DWG: 1
FAB DWG: 1
PCB DWG: 1
PROJECT
Wearable Wrist Device
Anders Bandt
PROJECT REVISION: 1.0
DOCUMENT REVISION: 1.0
DESIGN ITEM: 1.0
TITLE
Wearable Schematics
SIZE A3
CAGE CODE 58791
DWG NO. 1
REV 1
SCALE: 1:1
FILE NAME COMM_IMU.SchDoc
SHEET 2 OF 3



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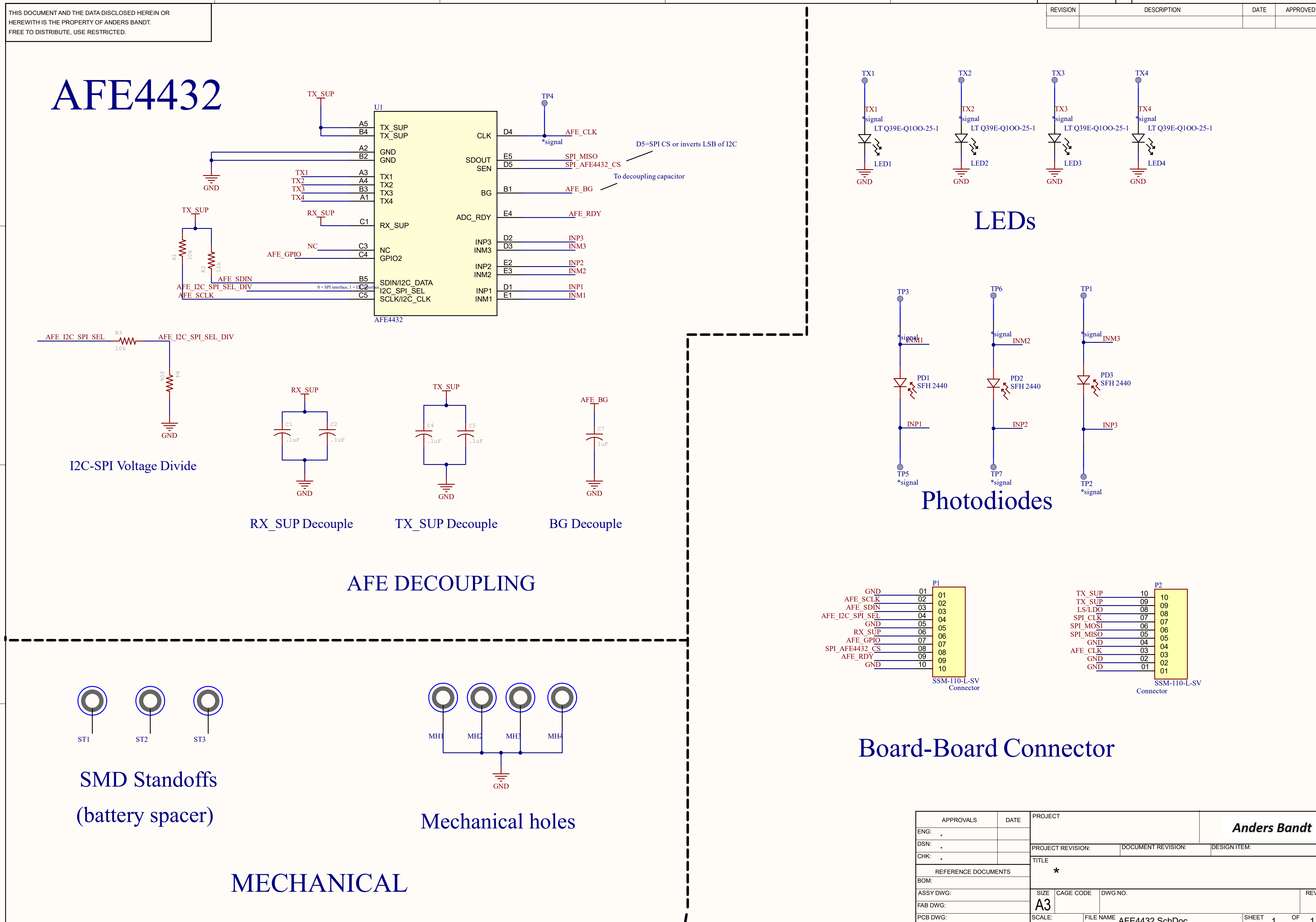
REVISION
DESCRIPTION
DATE
APPROVED
VDDS
VDDR
U7
VDDS
VDDS2
VDDS3
VDDR
VDDR_RF
VDDS_DCDC
DCDC_SW
Output from internal DC/DC converter
48M_OSC_P
48M_OSC_N
X48M_P
X48M_N
X32K_Q2
X32K_Q1
MCU_RESET
cJTAG_TCK
cJTAG_TMSC
RESET_N
JTAG_TCKC
JTAG_TMSC
CC2642R1FRGZR
DIO_0
DIO_1
DIO_2
DIO_3
DIO_4
DIO_5
DIO_6
DIO_7
DIO_8
DIO_9
DIO_10
DIO_11
DIO_12
DIO_13
DIO_14
DIO_15
DIO_16
DIO_17
DIO_18
DIO_19
DIO_20
DIO_21
DIO_22
DIO_23
DIO_24
DIO_25
DIO_26
DIO_27
DIO_28
DIO_29
DIO_30
RF_P
RF_N
DCOUPL
EGP
LED_STATUS
CBUS2
UART_RX
UART_TX
FT230X_RESET
AFE_CLK
IMU_INT2
IMU_INT1
SPI_MISO
SPI_MOSI
SPI_CLK
SPI_IMU_CS
JTAG_TDO
JTAG_TDI
BMS_CD
BMS_INT
BMS_ISET
I2C_SDA
BOOST_EN
I2C_SCL
AFE_I2C_SPI_SEL
AFE_GPIO
SPI_AFE4432_CS
AFE_RDY
RF_P
DIO2
DIO3
C26
1uF
GND
Pins DIO_23 through DIO_30 have analog capability
LED_STATUS
R17
100
LED2
STATUS LED
TP12
TP13
DCDC_SW
L4
10uH
VDDR
Pin 45
Pin 48
C29
10uF
C31
0.1uF
C33
0.1uF
VDDR Decoupling
MCU + Bluetooth
VDDS
Pin 44
Pin 22
Pin 13
Pin 34
C27
0.1uF
C28
0.1uF
C30
0.1uF
C32
10uF
C34
0.1uF
VDDS Decoupling
XTAL1
48 MHz oscillator
C24
15pF
C25
15pF
Crystal
48 MHz Oscillator
J10
OUT
IN
GND
GND
G1
G2
RF
C35
1pF
L5
3.9nH
L6
2.7nH
ANT1
RF Test Port
Antenna System
APPROVALS
DATE
ENG:
DSN:
CHK:
4/20/23
REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:
PROJECT
Wearable Wrist Device
Anders Bandt
PROJECT REVISION: 1.0
DOCUMENT REVISION: 1.0
DESIGN ITEM: 1.0
TITLE
Wearable Schematics
SIZE A3
CAGE CODE 58791
DWG NO. 1
REV 1
SCALE: 1:1
FILE NAME MCU.SchDoc
SHEET 3 OF 3



REVISION	DESCRIPTION	DATE	APPROVED

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT
Anders Bandt
PROJECT REVISION: DOCUMENT REVISION: DESIGN ITEM:
TITLE *
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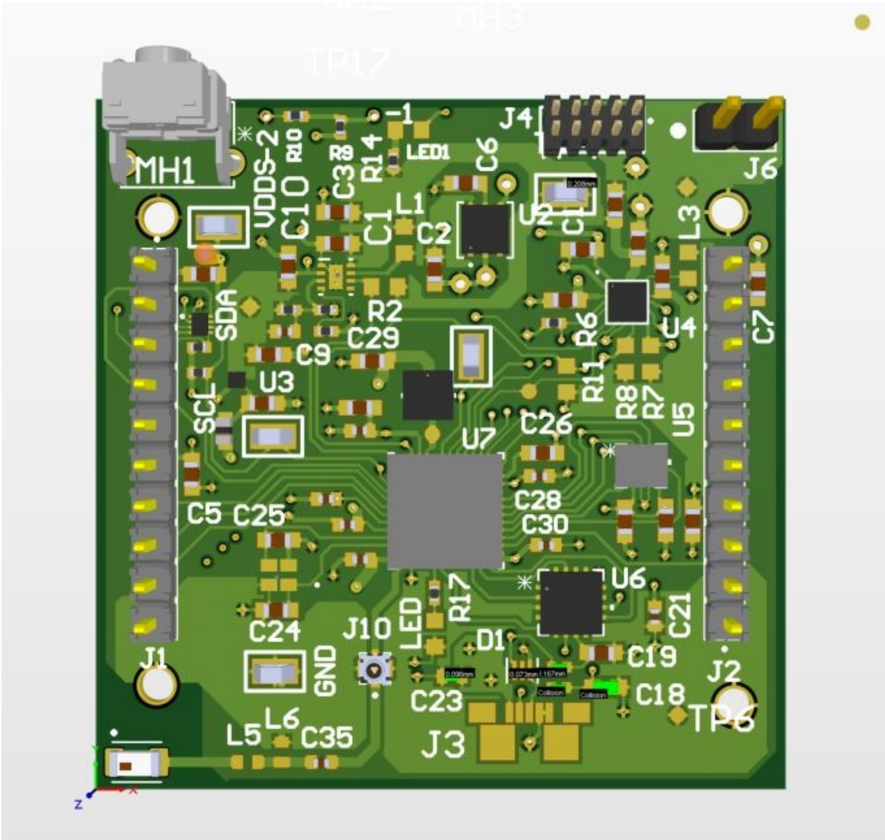
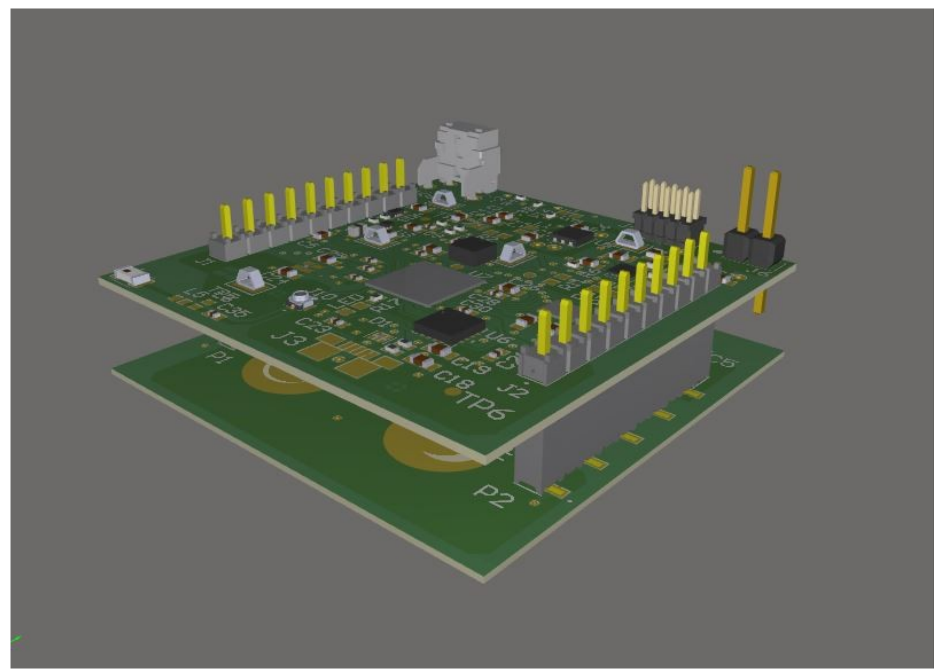
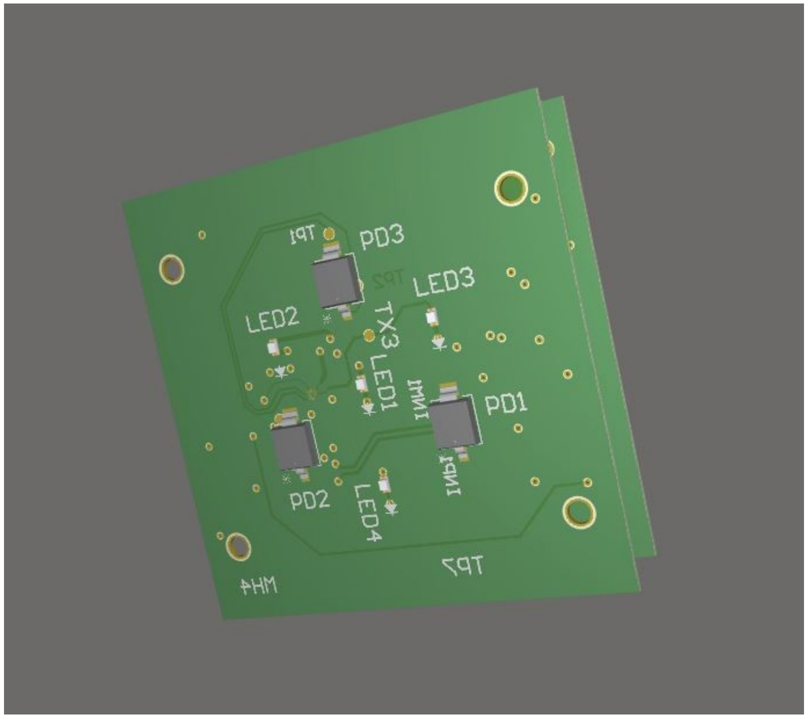
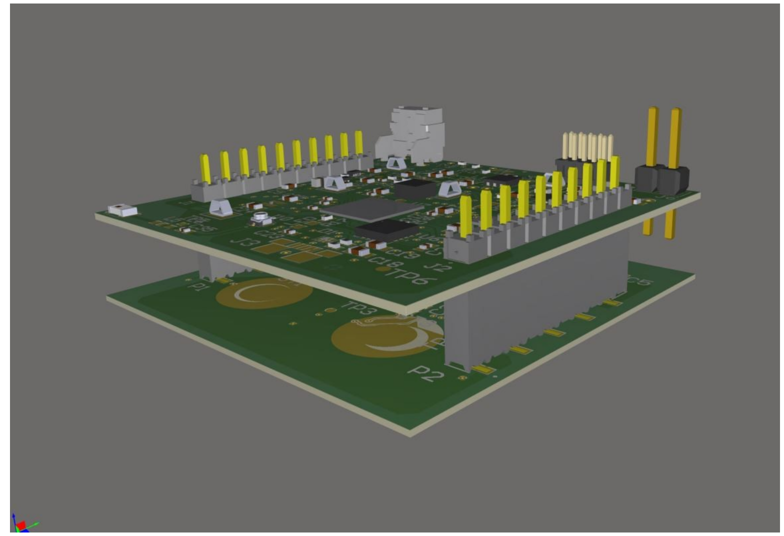
SCALE: FILE NAME AFE4432.SchDoc SHEET 1 OF 1

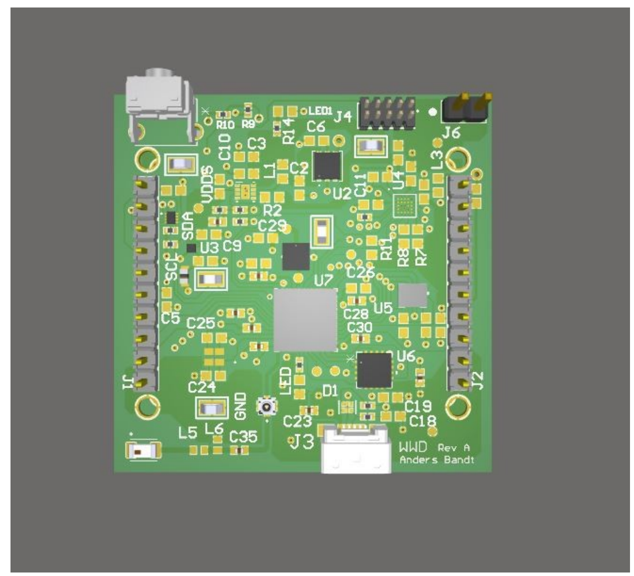


MH1
MH3
TP17
VDDS-2
R10
R9
C3
R14
LED1
C6
J4
J6
C10
C1
L1
C2
U2
L3
SDA
R2
C29
C9
R6
U4
C7
U3
SCL
R11
R8
R7
U5
C26
U7
C5
C25
C28
C30
U6
R17
C21
LED
C24
J10
D1
C19
J1
GND
J2
C23
C18
TP6
L5
L6
C35
J3

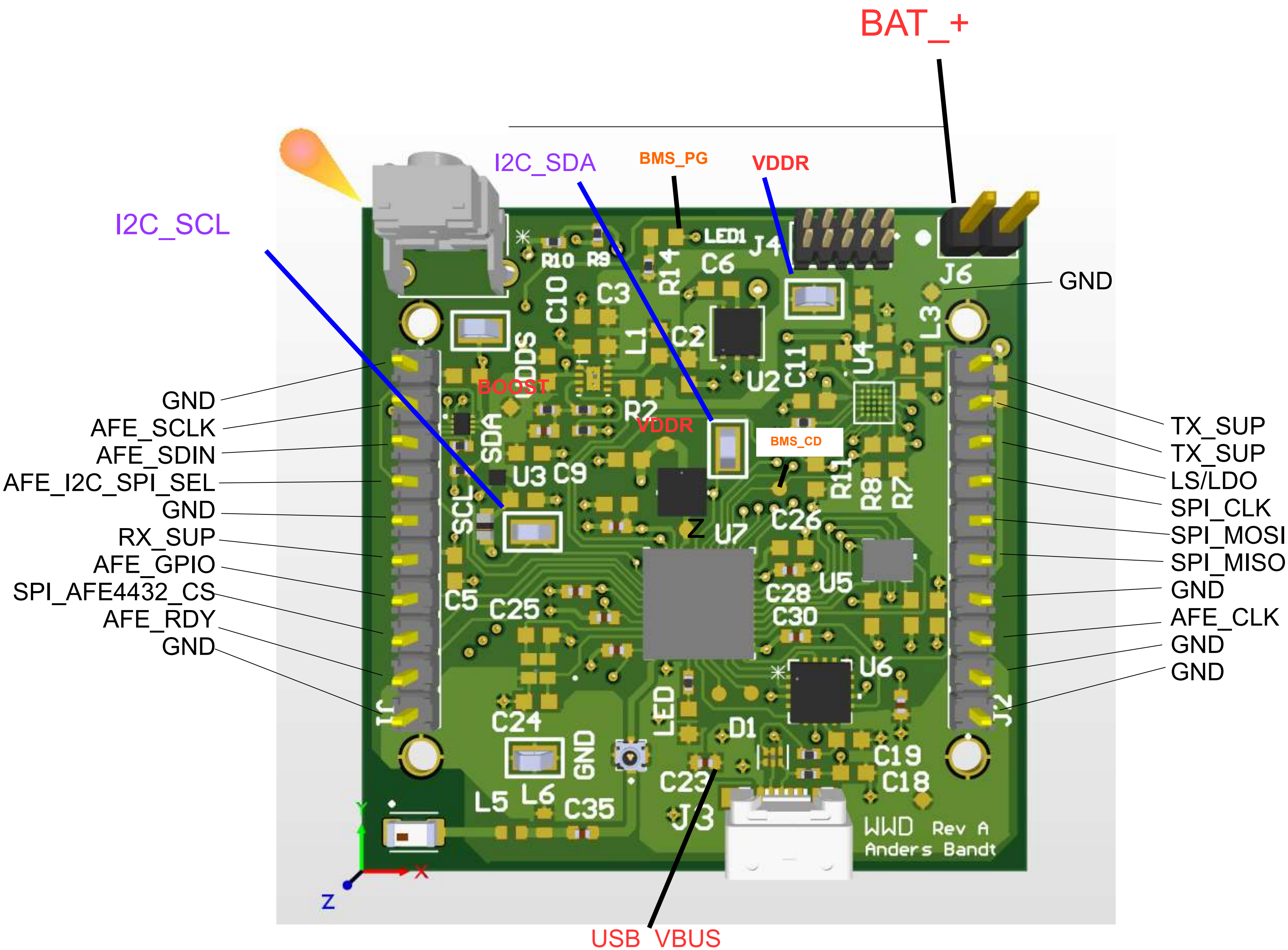


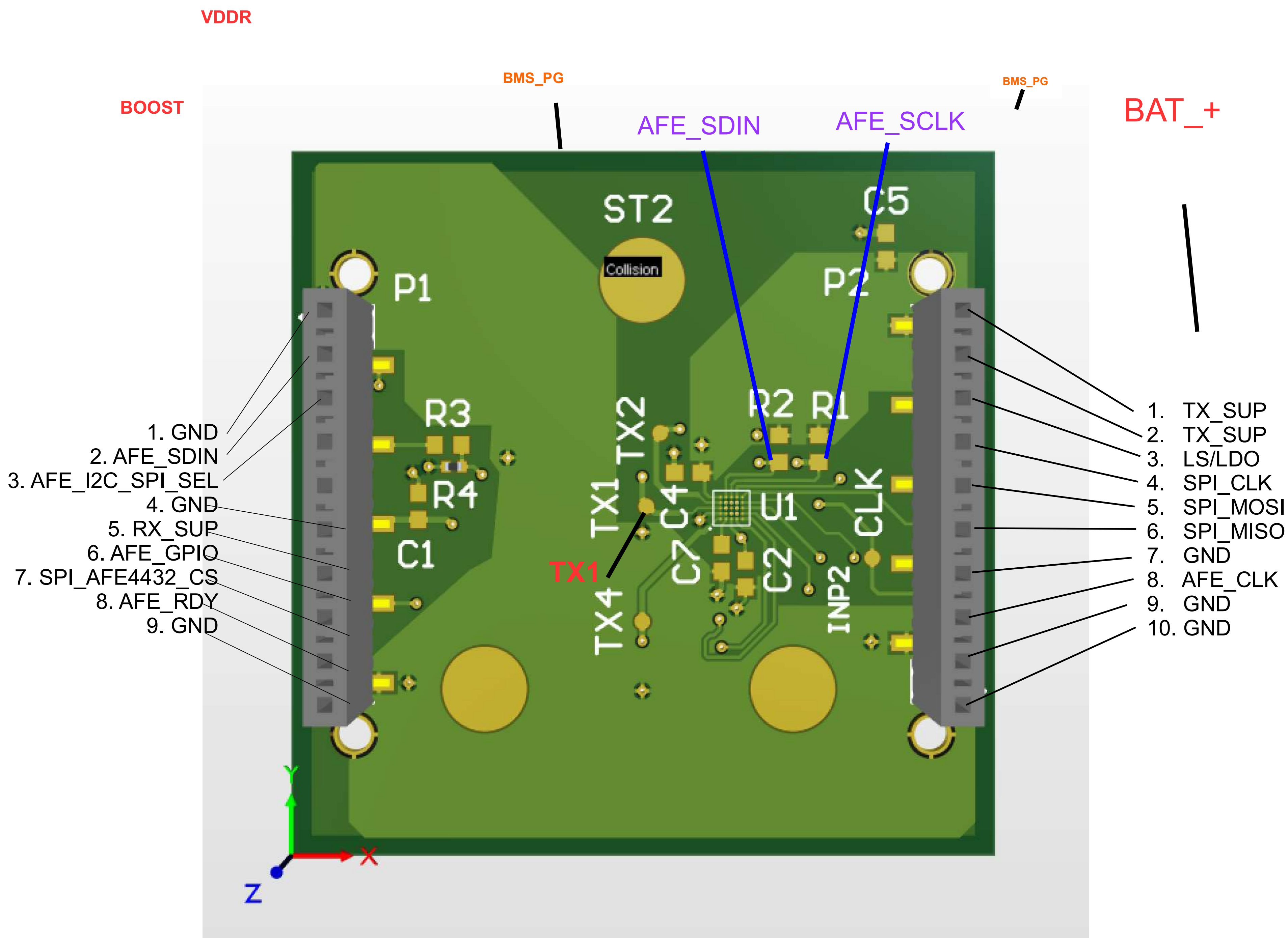

PD3
LED3
LED2
TX3
LED1
PD1
PD2
LED4

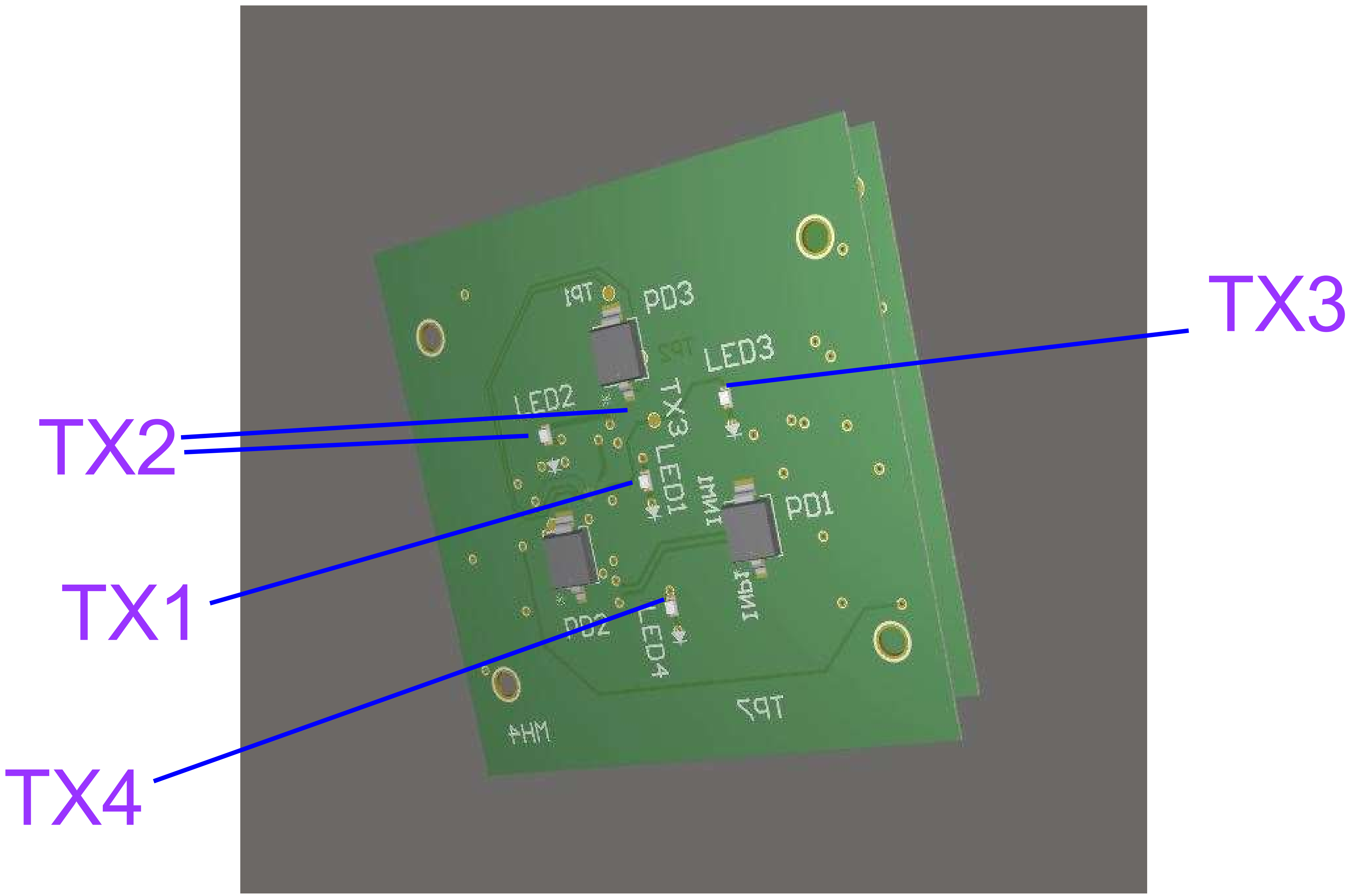

TP6
P2

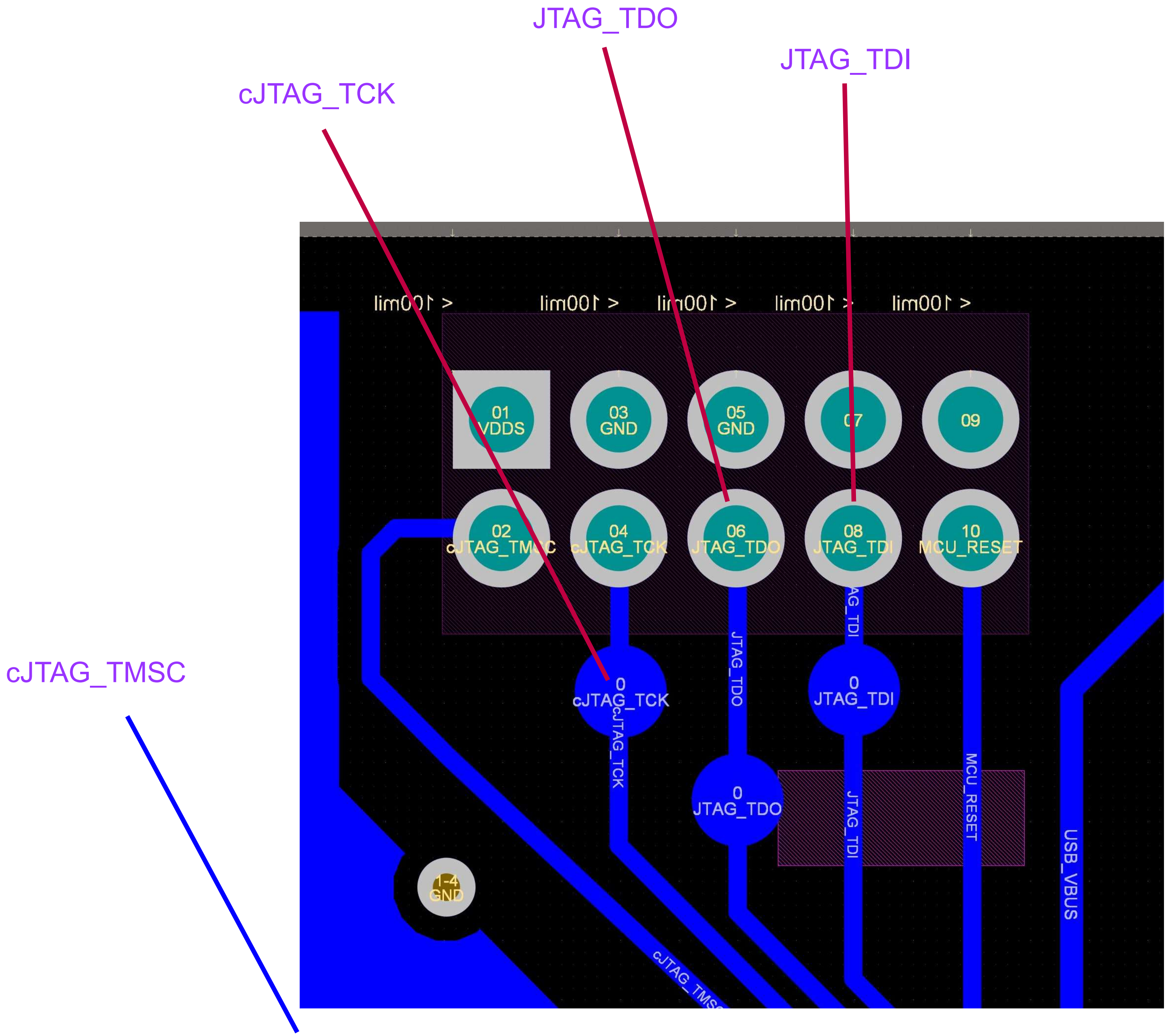

J4
J6
J1
J2
J3
U7
U6
WWD Rev A
Anders Bandt

	Ref. Des	Manufacturer	Quantity	Digikey Part #	Manufacter Part #	Area	Usage	General Description	Package	Datasheet	Ordered?	Notes
			1		AFE4432	Sensing	Analog front end				Y	
		OSRAM	4	475-3441-1-ND	LT Q39E-Q1OO-25-1	Sensing	PPG LED emitter	LED GREEN DIFFUSED 0603 SMD			Y	
		OSRAM	3	475-3521-1-ND	SFH 2440	Sensing	Photodiode	SENSOR PHOTODIODE 950NM 2SMD			Y	
			1	1428-ICM-42670-PCT-ND	ICM-42670-P	Sensing	IMU					
			1	535-13925-1-ND	ABM11W-48.0000MHZ-7-B1U-T3	Timing	48 MHz crystal (MCU)	CRYSTAL 48.0000MHZ 7PF SMD				
		TI	1		CC2642R1FRGZR	MCU	MCU				Y	
		Johanson Technology Inc.	1	712-2450AT18A0100001ECT-ND	2450AT18A0100001E	RF	2.4 GHZ, 50 ohm antenna	2.4GHz Chip RF Antenna 2.4GHz ~ 2.5GHz 0.5dBi Solder Surface Mount				
			1		MM8030-2610RK0	RF	RF test port	RF Connectors / Coaxial Connectors 50 OHM30V				
			1	490-1011-1-ND	BLM18AG121SN1D	Passive	Ferrite bead	FERRITE BEAD 120 OHM 0603 1LN				
			2	495-77047-1-ND	B82496C3229Z000	Passive, Power	2.2nH INDUCTOR (buck)	FIXED IND 2.2NH 1.5A 35 MOHM SMD				
			1	445-174641-1-ND	MHQ0402PSA2N7BT000	RF	T-match network (2.7nH)	FIXED IND 2.7NH 300MA 400MOHM SM	0402			
			1		LRC0402CS3N9GV001T	RF	T-match network (3.9nH)	FIXED IND 3.9NH 300MA 220MOHM SM	0402			
			1		74438335100	Passive, Power	10uH INDUCTOR (DC/DC)	FIXED IND 10UH 850MA 513MOHM SMD				
			1		C0402C109D4HAC7867	RF	T-match network (1pF)	CAP CER 1PF 16V X8R 0402				
			4		C0603C150K5RAC7867	Passive, Timing	15pF CAPACITOR	CAP CER 15PF 50V X7R 0603	0603			
			1	1276-1025-1-ND	CL05C101JB5NNNC	Passive	100pF CAPACITOR	CAP CER 100PF 50V C0G/NP0 0402	0402			
			1		0402YC103KAT2A	Passive	10nF CAPACITOR	CAP CER 10000PF 16V X7R 0402	0402			
			12		CL05A104KA5NNNC	Passive	0.1uF CAPACITOR	CAP CER 0.1UF 25V X5R 0402	0402			
			2		CL10A105KO8NNNC	Passive	1uF CAPACITOR	CAP CER 1UF 16V X5R 0603	0603			
			1		GRM155Z71A225KE44D	Passive	2.2uF CAPACITOR	CAP CER 2.2UF 10V X7R 0402	0402			
			8		CL10A106MQ8NNNC	Passive	10uF CAPACITOR	CAP CER 10UF 6.3V X5R 0603	0603			
			1		CL10A226MQ8NRNC	Passive	22uF CAPACITOR	CAP CER 22UF 6.3V X5R 0603	0603			
			2	2037-PFR05S-220-FNHTR-ND	PFR05S-220-FNH	Passive	22 RESISTOR	RES SMD 22 OHM 1% 1/16W 0402	0402			
			2		RR0510P-101-D	Passive	100 RESISTOR	RES SMD 100 OHM 0.5% 1/16W 0402	0402			
			1	YAG4297CT-ND	RT0402BRD07665RL	Passive	665 RESISTOR	RES SMD 665 OHM .1% 1/16W 0402	0402			
			1	311-9.53KLRCT-ND	RC0402FR-079K53L	Passive	9.53k RESISTOR	RES 9.53K OHM 1% 1/16W 0402	0402			
			2	311-10KJRCT-ND	RC0402JR-0710KL	Passive	10k RESISTOR	RES 10K OHM 5% 1/16W 0402	0402			
			5	A126331CT-ND	CRG0603J10K	Passive	10k RESISTOR	10k resistor, 0603	0603			
			1	311-11.3KLRCT-ND	RC0402FR-0711K3L	Passive	11.3k RESISTOR	RES 11.3K OHM 1% 1/16W 0402	0402			
			2	10-ERJ-U02F2002XCT-ND	ERJ-U02F2002X	Passive	20k RESISTOR	RES 20K OHM 1% 1/10W 0402 SMD	0402			
			1	311-100KHRCT-ND	RC0603FR-07100KL	Passive	100k RESISTOR	RES 100K OHM 1% 1/10W 0603	0402			
			1	1292-WR04X2203FTLCT-ND	WR04X2203FTL	Passive	220k RESISTOR (BOOST FB)	RES 220K OHM 1% 1/16W 0402	0402			
			1		RC0402FR-07698KL	Passive	698k RESISTOR (BOOST FB)	RES 698K OHM 1% 1/16W 0402	0402			
			2	160-1447-1-ND	LTST-C191KRKT	Interface, LED	Indicator LED	Red 631nm LED Indication - Discrete 2V 0603 (1608 Metric)	0603		Y	
			1		TS11-674-65-BK-160-RA-D	Interface	Reset button				Y	
			1		BQ25120AYFPT	Power	BMS SYSTEM				Y	
			1		NTCLE100E3103HT1	Power	Battery thermistor	NTC Thermistor 10k 3977K BEAD				
	U2	TI	1	LP38693SD-5.0/NOPBTR-ND	LP38693SD-5.0/NOPB	Power	TX LDO					
	U3	TI	1	LP5900TL-1.8/NOPBTR-ND	LP5900TL-1.8/NOPB	Power	RX LDO	IC LDO, 1.8V, 150mA				
			1	296-39439-2-ND	TPS61252DSGR	Power	BOOST	IC BOOST CONV ADJ 1.5A 8WSON				
			1		RMCF0402ZT0R00	Jumper	0 OHM JUMPER	RES 0 OHM JUMPER 1/16W 0402	0402			
			1		77311-118-02LF	Connector, Power	Battery power clip	CONN HEADER VERT 2POS 2.54MM				
			1	PCA9306FMUTAGOSCT-ND	PCA9306FMUTAG	Comm, IC	I2C voltage translator					
	J1, J2		2		TSW-110-14-F-S	Connector	Board to board	Through hole, 10 position, 2.54mm				
	P1, P2		2		SSM-110-L-SV	Connector	Board to board	Surface mount, 10 position, 2.54mm				
			1		FTSH-105-01-F-D	Connector, Comm	JTAG debug input					
			1		GMCBA058211230HHR	Comm, Power, Connector	USB Micro-B input					
			1	296-21884-2-ND	TPD2E001DRYR	EMC, Connector	ESD suppression (USB-C)					
		FTDI	1	768-1153-ND	FT230XQ-R	Comm					Y	
			5	36-5017CT-ND	5017	Test	Test Point					
			3	36-4892CT-ND	4892	Mechanical	Battery spacer	Round Standoff Threaded #6-32 Steel 0.125" (3.18mm) 1/8"				
							Standoff					
		CABLE 138 RF-0100-A-2	1	2072-CABLE138RF-0100-A-2-ND		RF	U.Fl to Coax connector					

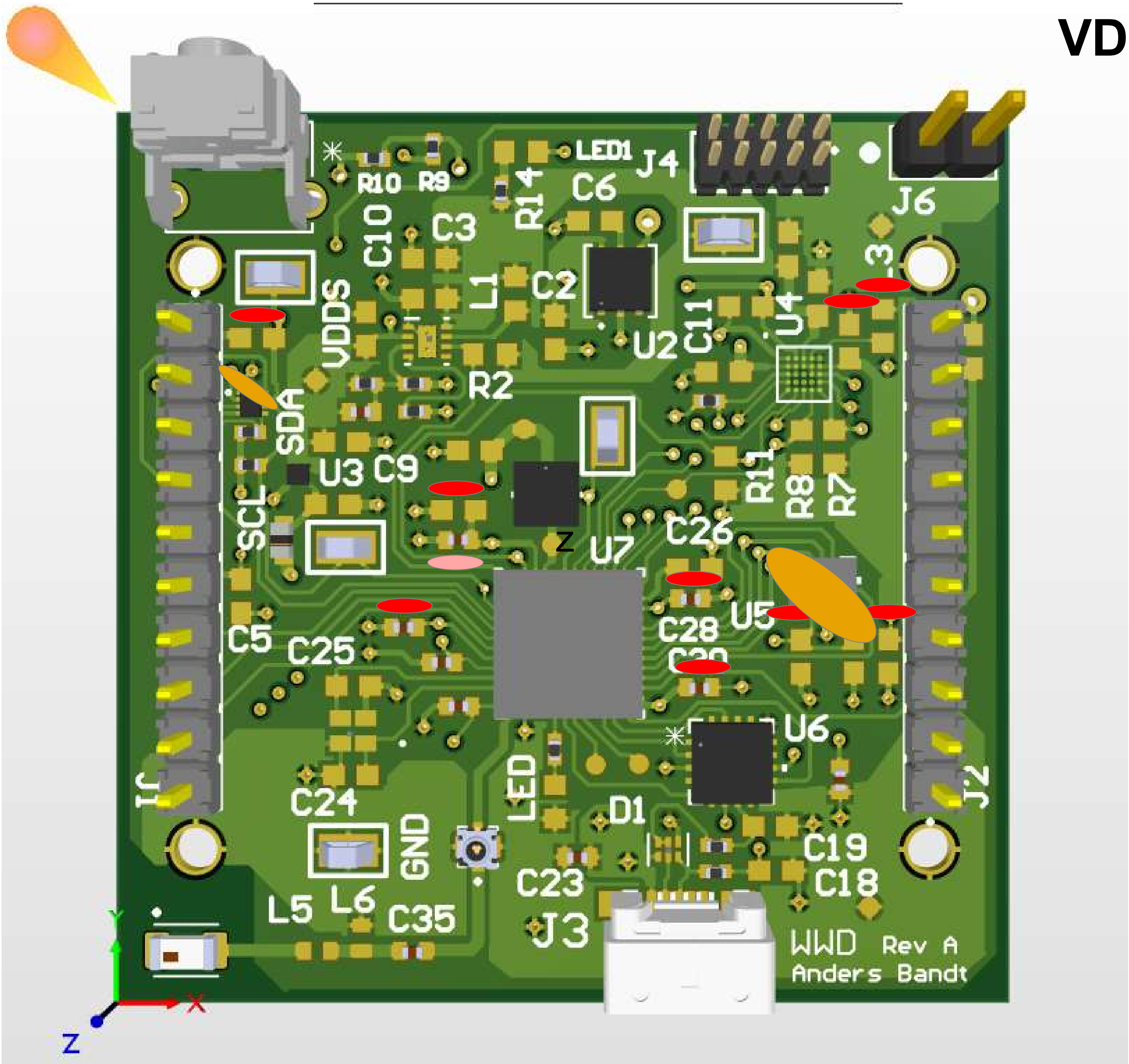

BAT_+
I2C_SCL
I2C_SDA
BMS_PG
VDDR
GND
BOOST
VDDR
BMS_CD
GND
AFE_SCLK
AFE_SDIN
AFE_I2C_SPI_SEL
GND
RX_SUP
AFE_GPIO
SPI_AFE4432_CS
AFE_RDY
GND
TX_SUP
TX_SUP
LS/LDO
SPI_CLK
SPI_MOSI
SPI_MISO
GND
AFE_CLK
GND
GND
USB_VBUS
WWD Rev A
Anders Bandt


VDDR
BOOST
BMS_PG
BMS_PG
AFE_SDIN
AFE_SCLK
BAT_+
ST2
C5
P1
P2
Collision
R3
R4
C1
R2
R1
TX2
TX1
C4
U1
CLK
C7
C2
INP2
TX4
TX1
1. GND
2. AFE_SDIN
3. AFE_I2C_SPI_SEL
4. GND
5. RX_SUP
6. AFE_GPIO
7. SPI_AFE4432_CS
8. AFE_RDY
9. GND
1. TX_SUP
2. TX_SUP
3. LS/LDO
4. SPI_CLK
5. SPI_MOSI
6. SPI_MISO
7. GND
8. AFE_CLK
9. GND
10. GND
Y
X
Z


TX3
TX2
TX1
TX4
PD3
LED3
LED2
TX3
LED1
PD1
PD2
LED4


JTAG_TDO
JTAG_TDI
cJTAG_TCK
cJTAG_TMSC
01 VDDS
03 GND
05 GND
07
09
02 cJTAG_TMSC
04 cJTAG_TCK
06 JTAG_TDO
08 JTAG_TDI
10 MCU_RESET
0 cJTAG_TCK
0 JTAG_TDI
0 JTAG_TDO
1-4 GND
MCU_RESET
USB_VBUS

VDDS SHORTS



= IC short

= top

= bottom